



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SANGAMESWARI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

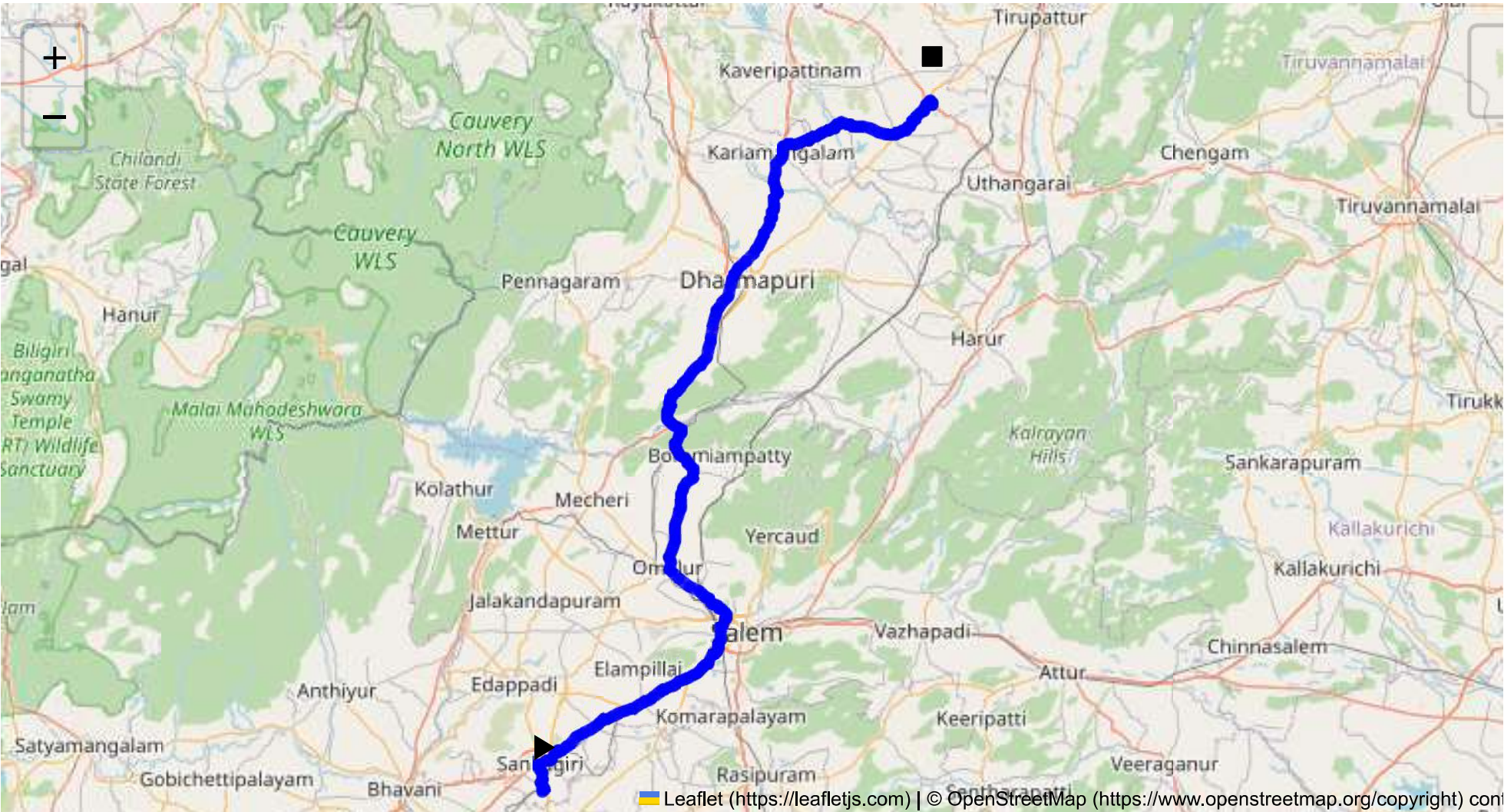
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 154.61 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.7 hours
Start: (11.4381, 77.8734)
End: (12.377933, 78.412164)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map

The route from Sangagiri to Batrahalli via Suramangalam, Salem, and Krishnagiri spans approximately 154.61 kilometers. The journey typically takes around 3 hours for heavy vehicles transporting hazardous

material, following NH44 which is a major highway known for connecting various regions across Tamil Nadu.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Conditions:** The route passes through regions with a tropical climate; hot and humid during summer months (March to June), and tempered by the monsoon rains from June to September.
- **Potential Hazards:** Heavy rains during the monsoon can lead to waterlogging and reduced visibility. Occasionally, there may be landslides or flooding, particularly near hilly sections.

3. Traffic Patterns

- **Peak Hours:** The route experiences peak traffic typically between 7-10 AM and 5-8 PM, particularly near major towns such as Salem and Krishnagiri.
- **Congestion-Prone Areas:** Significant congestion is often reported near Salem city and Krishnagiri junctions, where traffic narrows due to urban intersections.

4. Assessment of Road Quality and Infrastructure

- The NH44 highway generally offers a well-maintained roadway suitable for heavy vehicles. However, certain segments near urban areas might have potholes and uneven surfaces.
- Some parts of the road near market areas may be narrower, posing challenges for larger, heavy vehicles navigating tight turns or crowded streets.

5. Suggestions for Alternative Routes

- Emergency routes might include state highways branching off NH44, bypassing certain congested towns. Local government-issued maps often highlight these alternatives during adverse conditions like road repairs or accidents.

6. Local Regulations Affecting Hazardous Material Transport

- The transport of hazardous materials is subject to strict regulations mandating specific travel times (often during non-peak hours), designated routes, and required permits. Violations may incur significant penalties.

7. Historical Incidents

- While the route is typically safe for heavy vehicles, there have been occasional accidents attributed to reckless driving or fatigue. Keeping abreast of local news channels can provide updates on route safety.

8. Environmental Considerations and Sensitive Areas

- Many sections pass through agricultural regions, necessitating caution with hazardous materials. Drivers must be vigilant in sensitive areas to prevent environmental contamination in case of spills.
- There are also some wildlife crossings, for which extra caution is advised to prevent accidents.

9. Analysis of Communication Coverage

- While most of the route generally enjoys good mobile and radio coverage, there could be some dead zones near remote sections, especially between towns and villages.

10. Estimated Emergency Response Times

- Near urban centers like Salem or Krishnagiri, emergency response times are typically fast, within 15-30 minutes.
- In more rural and remote stretches, response times could extend to over an hour.

12. Overall Summary of Risk Assessment

- **Overall Rating:** Moderate Risk
- **Key Concerns:** Weather-related hazards during the monsoon season, congestion in urban centers, and rural sections lacking robust emergency support.
- **Recommendations:** Familiarize with alternate routes, adhere to local regulations, ensure communication tools are well maintained, prepare for weather influences, and maintain vehicles to mitigate breakdown risks.

By implementing best practices in vehicle operation and being prepared for environmental and infrastructural challenges, the overall risks associated with this route can be effectively managed for the safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	High	11.47403, 77.86841	15 KM/Hr
6	Turn	Medium	11.47986, 77.88397	30 KM/Hr
7	Turn	High	11.49149, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	High	11.49224, 77.89601	15 KM/Hr
9	Turn	Medium	11.53501, 77.95707	30 KM/Hr
10	Turn	Medium	11.53601, 77.95856	30 KM/Hr
11	Turn	Medium	11.62403, 78.10893	30 KM/Hr
12	Turn	Medium	11.62917, 78.11648	30 KM/Hr
13	Turn	High	11.64180, 78.11808	15 KM/Hr
14	Turn	High	11.64325, 78.11993	15 KM/Hr
15	Turn	Medium	11.67865, 78.12803	30 KM/Hr
16	Turn	Medium	11.68201, 78.12639	30 KM/Hr
17	Turn	Medium	11.69519, 78.10775	30 KM/Hr
18	Turn	Medium	11.73958, 78.05105	30 KM/Hr
19	Turn	Medium	11.75835, 78.05016	30 KM/Hr
20	Turn	Medium	11.86847, 78.08367	30 KM/Hr
21	Turn	Medium	11.87942, 78.08197	30 KM/Hr
22	Turn	Medium	11.92585, 78.06507	30 KM/Hr
23	Turn	Medium	11.93319, 78.06419	30 KM/Hr
24	Turn	Medium	11.93557, 78.06103	30 KM/Hr
25	Turn	Medium	11.93704, 78.05402	30 KM/Hr
26	Turn	High	11.95818, 78.04978	15 KM/Hr
27	Turn	Medium	11.95988, 78.04948	30 KM/Hr
28	Turn	Medium	11.96304, 78.05136	30 KM/Hr
29	Turn	High	11.96399, 78.05135	15 KM/Hr
30	Turn	High	11.96580, 78.04923	15 KM/Hr
31	Turn	Medium	11.96675, 78.04939	30 KM/Hr
32	Turn	Medium	11.96791, 78.05070	30 KM/Hr
33	Turn	Medium	12.16983, 78.16769	30 KM/Hr
34	Turn	Medium	12.19881, 78.18157	30 KM/Hr
35	Turn	Medium	12.22459, 78.19136	30 KM/Hr
36	Turn	Medium	12.24731, 78.19601	30 KM/Hr
37	Turn	Medium	12.25060, 78.19977	30 KM/Hr
38	Turn	Medium	12.25329, 78.20039	30 KM/Hr
39	Turn	Medium	12.29110, 78.19909	30 KM/Hr
40	Turn	Medium	12.31483, 78.20789	30 KM/Hr
41	Turn	High	12.31904, 78.21608	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
42	Turn	Medium	12.31827, 78.21725	30 KM/Hr
43	Turn	Medium	12.32435, 78.24037	30 KM/Hr
44	Turn	High	12.32373, 78.24193	15 KM/Hr
45	Turn	Medium	12.34226, 78.28138	30 KM/Hr
46	Turn	Medium	12.34566, 78.28283	30 KM/Hr
47	Turn	Medium	12.34341, 78.30518	30 KM/Hr
48	Turn	Medium	12.33200, 78.35662	30 KM/Hr
49	Turn	Medium	12.33289, 78.35872	30 KM/Hr
50	Turn	High	12.33199, 78.36425	15 KM/Hr
51	Turn	Medium	12.34340, 78.38518	30 KM/Hr
52	Turn	Medium	12.35074, 78.38809	30 KM/Hr
53	Turn	High	12.37538, 78.40978	15 KM/Hr
54	Blind Spot	Blind Spot	12.37317, 78.41475	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
11	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
13	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
18	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
19	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
20	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
21	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
23	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
24	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
25	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
26	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
27	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
30	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
31	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
32	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
33	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
36	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
39	fire_station	Fire Station	12.3328308, 78.3489464	30 km/h	Medium
40	hospital	Government Hospital, Pochampalli	12.3322477, 78.365767	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
12	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
16	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
17	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
22	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
28	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
29	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
35	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
37	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
38	school	Chellampatti -PUPS	12.3467776, 78.2947171	30 km/h	Medium
41	school	GUNA Matric Higher Secondary School	12.3722333, 78.4086433	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47986, 77.88397



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62917, 78.11648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67865, 78.12803



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68201, 78.12639



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73958, 78.05105



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75835, 78.05016



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93319, 78.06419



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93557, 78.06103



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.16983, 78.16769



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24731, 78.19601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31483, 78.20789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31904, 78.21608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.31827, 78.21725



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32435, 78.24037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.32373, 78.24193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.34341, 78.30518



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.33200, 78.35662



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.33289, 78.35872



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.33199, 78.36425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.34340, 78.38518



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.35074, 78.38809



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.37538, 78.40978



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.37317, 78.41475

